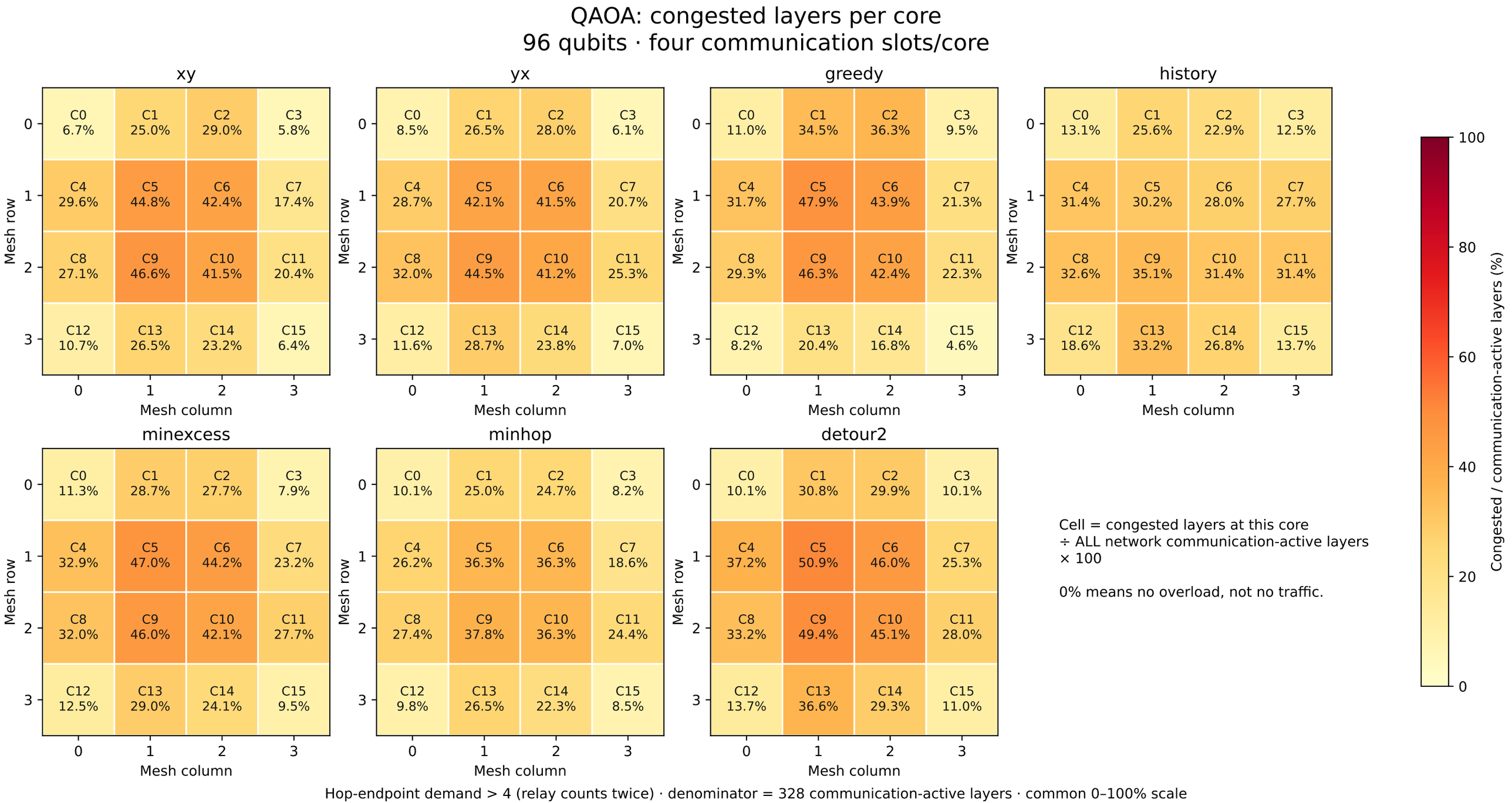
QAOA: congested layers per core
96 qubits · four communication slots/core
xy
C0 6.7%
C1 25.0%
C2 29.0%
C3 5.8%
C4 29.6%
C5 44.8%
C6 42.4%
C7 17.4%
C8 27.1%
C9 46.6%
C10 41.5%
C11 20.4%
C12 10.7%
C13 26.5%
C14 23.2%
C15 6.4%
yx
C0 8.5%
C1 26.5%
C2 28.0%
C3 6.1%
C4 28.7%
C5 42.1%
C6 41.5%
C7 20.7%
C8 32.0%
C9 44.5%
C10 41.2%
C11 25.3%
C12 11.6%
C13 28.7%
C14 23.8%
C15 7.0%
greedy
C0 11.0%
C1 34.5%
C2 36.3%
C3 9.5%
C4 31.7%
C5 47.9%
C6 43.9%
C7 21.3%
C8 29.3%
C9 46.3%
C10 42.4%
C11 22.3%
C12 8.2%
C13 20.4%
C14 16.8%
C15 4.6%
history
C0 13.1%
C1 25.6%
C2 22.9%
C3 12.5%
C4 31.4%
C5 30.2%
C6 28.0%
C7 27.7%
C8 32.6%
C9 35.1%
C10 31.4%
C11 31.4%
C12 18.6%
C13 33.2%
C14 26.8%
C15 13.7%
minexcess
C0 11.3%
C1 28.7%
C2 27.7%
C3 7.9%
C4 32.9%
C5 47.0%
C6 44.2%
C7 23.2%
C8 32.0%
C9 46.0%
C10 42.1%
C11 27.7%
C12 12.5%
C13 29.0%
C14 24.1%
C15 9.5%
minhop
C0 10.1%
C1 25.0%
C2 24.7%
C3 8.2%
C4 26.2%
C5 36.3%
C6 36.3%
C7 18.6%
C8 27.4%
C9 37.8%
C10 36.3%
C11 24.4%
C12 9.8%
C13 26.5%
C14 22.3%
C15 8.5%
detour2
C0 10.1%
C1 30.8%
C2 29.9%
C3 10.1%
C4 37.2%
C5 50.9%
C6 46.0%
C7 25.3%
C8 33.2%
C9 49.4%
C10 45.1%
C11 28.0%
C12 13.7%
C13 36.6%
C14 29.3%
C15 11.0%
Mesh row
Mesh column
Cell = congested layers at this core
÷ ALL network communication-active layers
× 100
0% means no overload, not no traffic.
Congested / communication-active layers (%)
0
20
40
60
80
100
Hop-endpoint demand > 4 (relay counts twice) · denominator = 328 communication-active layers · common 0–100% scale